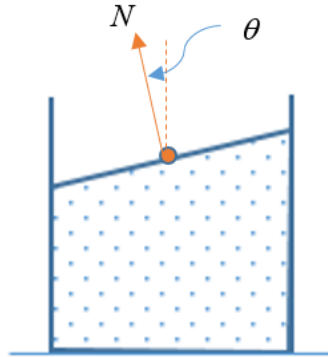


Q5



Option A: Consider a molecule on the water surface. The surrounding water exerts a normal force N on the molecule. For horizontal motion, using N2L, $N\sin\theta = ma$ where m is the mass of the molecule.

Option B is for the case of the cup of water decelerating when moving towards the left.

Option C is when the cup of water is stationary or moving at constant velocity.

Option D is for the case where the cup is being spun in circular motion about an axis left of the cup.

Ans: A

Q6